



Lista2



A partir da equação governante do estado quase-geostrofico da atmosfera.

$$\frac{\partial \nabla^2 \psi}{\partial t} - \frac{f_0^2}{gh_0} \frac{\partial \psi}{\partial t} = - \frac{\partial \psi}{\partial x} \left(\frac{\partial \nabla^2 \psi}{\partial y} - \frac{f_0^2}{gh_0} \frac{\partial \psi}{\partial y} \right) + \frac{\partial \psi}{\partial y} \left(\frac{\partial \nabla^2 \psi}{\partial x} - \frac{f_0^2}{gh_0} \frac{\partial \psi}{\partial x} \right) - \beta \frac{\partial \psi}{\partial x}$$

3 non-divergent shallow water equations

(QG-)vorticity equation (non divergent):

$$\frac{\partial \zeta_g}{\partial t} + u_g \frac{\partial \zeta_g}{\partial x} + v_g \frac{\partial \zeta_g}{\partial y} + \beta v_g = 0$$

Or:

$$\begin{aligned} \frac{\partial \nabla^2 \Psi}{\partial t} &= - \frac{\partial \Psi}{\partial x} \frac{\partial \nabla^2 \Psi}{\partial y} + \frac{\partial \Psi}{\partial y} \frac{\partial \nabla^2 \Psi}{\partial x} - \beta \frac{\partial \Psi}{\partial x} \\ &= -J(\Psi, \nabla^2 \Psi) - \beta \frac{\partial \Psi}{\partial x} \end{aligned}$$



Atmosphere Numerical Modeling Group

1) Implemente este modelo baseado na Equação Barotrópica e calcule a função de corrente ψ

https://github.com/pkubota/pgmet_2025/tree/main/MET-348-3/baro_model_forecast

2) Integre o modelo para o caso do furacão Catarina e compare os campos de velocidade do vento com os dados do Era5?

Link dos dados do furacão Catarina

https://ftp.cptec.inpe.br/pesquisa/bam/paulo.kubota/externo/MET-348-3/scm_pgmet_2025.tar.gz

https://ftp.cptec.inpe.br/pesquisa/bam/paulo.kubota/externo/MET-348-3/baro_model_forecast/dataout/

